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$$(1+i)z^2 + (3-7i)z - 10 = 0$$

$$D = (3-7i)^2 - 4 \cdot (1+i) \cdot (-10)$$

$$= 9 - 42i + 49i^2 + 40 + 40i$$

$$= 9 - 42i - 49 + 40 + 40i$$

$$= -2i$$

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \end{cases}$$

$$\begin{cases} x^2 = y^2 \\ xy = -1 \end{cases}$$

De resolvente vierkantsvergelijking:

$$\lambda^2 - 1 = 0$$

$$\lambda_{1,2} = \pm 1$$

$$x = 1 \text{ en } y = -1$$

$$b < 0:$$

$$\sqrt{D} = (1-i) \text{ of } D = (-1+i)$$

Oplossingen voor  $z$ :

$$z_{1,2} = \frac{-(3-7i) \pm (1-i)}{2(1+i)}$$

$$z_1 = 1 + 2i$$

$$z_2 = 1 + 3i$$

## References